

Wave Forces on Structures: Goda and Morison

Two standard methods: the Goda formulae for wave pressure and force on a vertical wall or caisson, and the Morison equation for drag and inertia force on a slender pile. Companion note to the online calculator.

1. Vertical walls — the Goda method

For an upright (caisson) breakwater the wave pressure is idealised as a trapezoid acting on the wall. Using the design wave $H_{\max} \approx 1.8 \cdot H_s$ (limited by depth), the pressure at still-water level and the elevation up to which pressure acts are:

$$p_1 = \frac{1}{2}(1+\cos\beta)(\alpha_1 + \alpha_2 \cos^2\beta)\rho g H_{\max}, \quad \eta^* = 0.75(1+\cos\beta)H_{\max}, \quad p_3 = \alpha_3 p_1, \quad p_u = \frac{1}{2}(1+\cos\beta)\alpha_1 \alpha_3 \rho g H_{\max}.$$

p_1 acts at SWL, p_3 at the wall base and p_4 at the crest ($p_4 = p_1(1-h_c/\eta^*)$); p_u is the uplift under the base. The dimensionless coefficients capture depth and mound effects:

$$\alpha_1 = 0.6 + \frac{1}{2}[(4\pi h/L)/\sinh(4\pi h/L)]^2, \quad \alpha_2 = \min\{[(h_b-d)/3h_b](H_{\max}/d)^2, 2d/H_{\max}\}, \quad \alpha_3 = 1 - (h'/h)[1 - 1/\cosh(2\pi h/L)].$$

The horizontal force P is the area of the pressure trapezoid over the wall height, and the uplift U is the triangle under the base; both feed the sliding and overturning checks of the caisson.

2. Slender piles — the Morison equation

For a pile whose diameter is small compared with the wavelength ($D/L < \sim 0.2$), the in-line force per unit length is the sum of a velocity-dependent drag term and an acceleration-dependent inertia term:

$$f = \frac{1}{2}\rho C_D D \cdot u|u| + \rho C_m (\pi D^2/4) \cdot (du/dt),$$

with the horizontal velocity u and acceleration du/dt from linear theory. Drag and inertia are 90° out of phase, so the peak total force is obtained by combining their depth-integrated maxima:

$$F_{\max} = F_D + F_I^2/(4F_D) \text{ if } F_I \leq 2F_D, \text{ otherwise } F_{\max} = F_I. \quad \text{The Keulegan-Carpenter number } KC = u_{\max} T/D \text{ indicates whether inertia } (KC < 10) \text{ or drag } (KC > 20) \text{ dominates.}$$

3. Pressure diagram and force phasing

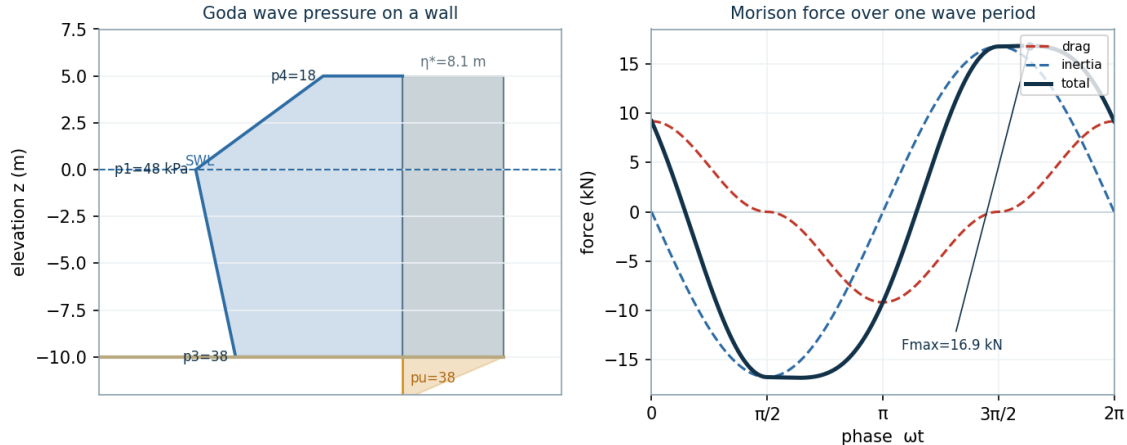


Figure 1. Left: Goda wave-pressure trapezoid on a vertical wall (worked example). Right: the Morison force over one wave period — drag and inertia (dashed) combine to a peak between their individual maxima (worked example).

4. Worked examples

Goda — vertical wall	Value
Inputs	$H_s=3$ m, $T=10$ s, $h=10$ m, $d=10$ m, $h_c=5$ m, $\beta=0$
Design height / wavelength	$H_{\max}=5.40$ m, $L=92.4$ m
$\alpha_1 / \alpha_2 / \alpha_3$	0.88 / 0.00 / 0.81
$p_1 / p_3 / p_4 / p_u$	47.7 / 38.5 / 18.3 / 38.5 kPa
Horizontal force, P	596 kN/m
Uplift, U ($B=8$ m) / overturning M	154 kN/m / 4234 kN·m/m
Morison — slender pile	Value

Inputs	H=3 m, T=8 s, h=10 m, D=1.0 m, $C_d=1.0$, $C_m=2.0$
Wavelength / D/L	70.9 m / 0.014
Surface velocity / KC	1.66 m/s / 13.3
Max drag / max inertia	9.2 kN / 16.8 kN
Max total force, $F_{\max}$	16.9 kN
Max moment about base	90 kN·m

Notation

H_s significant / H wave height; $H_{\max}$ design height; T period; h depth; h' wall-base depth; d berm depth; h_b depth $5H_s$ seaward; h_c crest above SWL; β wave angle; α_1 – α_3 Goda coefficients; p_1, p_3, p_4 pressures; p_u uplift; P horizontal force; U uplift force; D pile diameter; C_d, C_m drag/inertia coefficients; $u, du/dt$ velocity/acceleration; KC Keulegan-Carpenter number; ρ water density; g gravity.

References

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